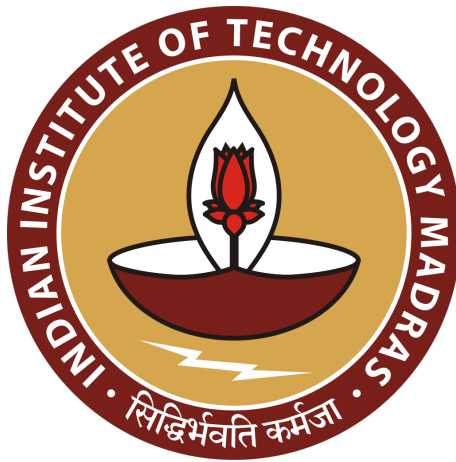


Assignment -3

Clustering



BT3041 - Analysis and Interpretation of Biological Data

Even Semester 2026

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NOTE: Clustering Algorithms are sensitive to the scale of the features, as such normalization/scaling of the feature data was first executed before beginning clustering.

Question 1

Agglomerative Hierarchical Clustering using Ward linkage was performed and was cut-off at maximum of 3 clusters. The results obtained are shown below:

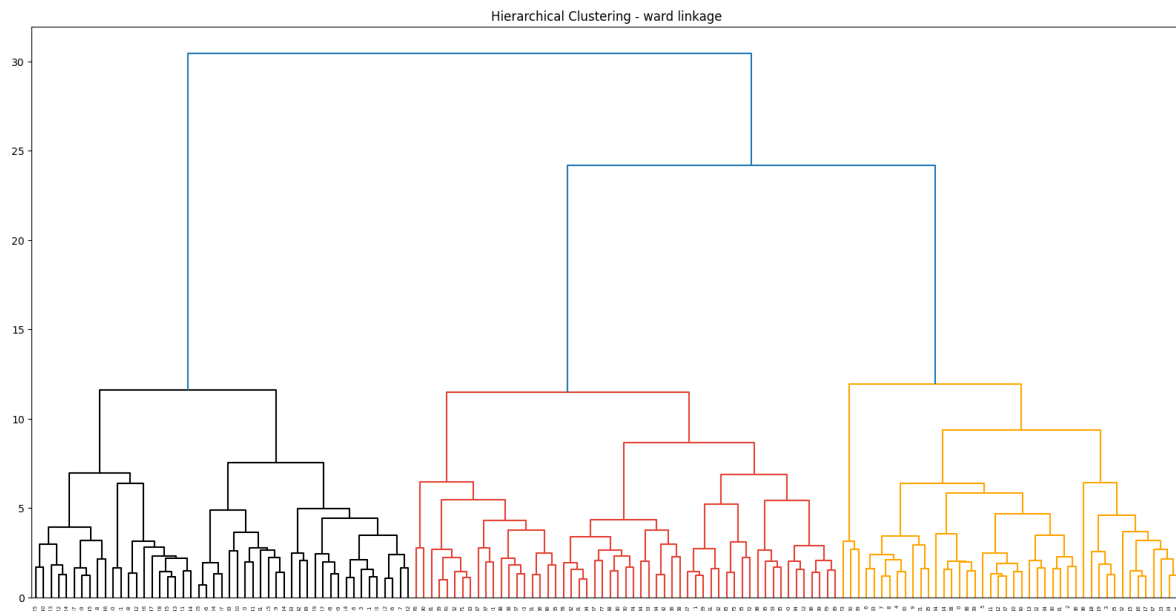


Figure 1.1: Agglomerative Hierarchical Clustering (Dendrogram) - Ward Linkage

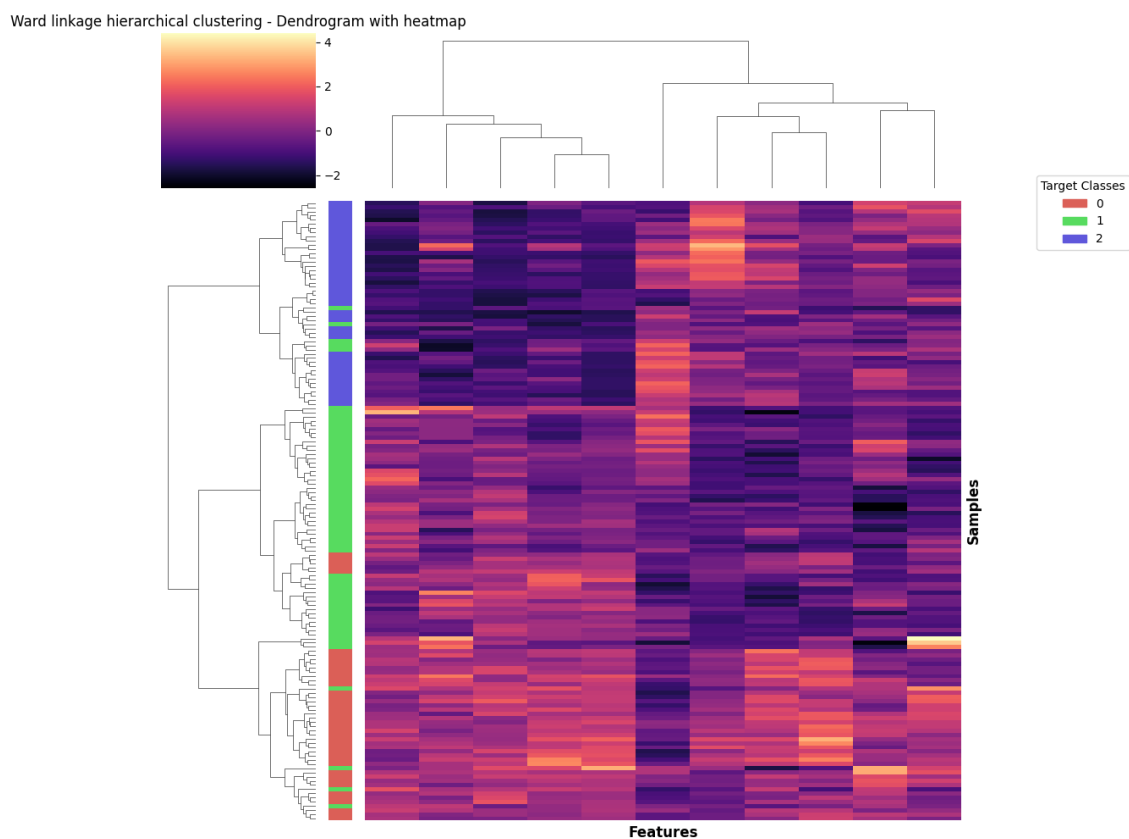


Figure 1.2: Ward linkage Agglomerative Hierarchical Clustering - Dendrogram with Heatmap

The quality of clustering is evaluated using the metric scores given below:

- Adjusted Rand Index (ARI) - 0.6808
- Adjusted Mutual Information (AMI) - 0.6664

From Figure 1.1, we see that the largest distance between any 2 samples in a groups is ≈ 13 .

From Figure 1.2, it is observed that clusters corresponding to groups 0 and 2 align well with the true labels, with distinct patterns seen in the heatmap across features.

From Figure 1.2, group 1 has several outliers, as some of its samples cluster with group 0 and others with group 2. This indicates that this intermediary group is highly sensitive and susceptible to noise.

Question 2

Single Linkage

Euclidean Distance

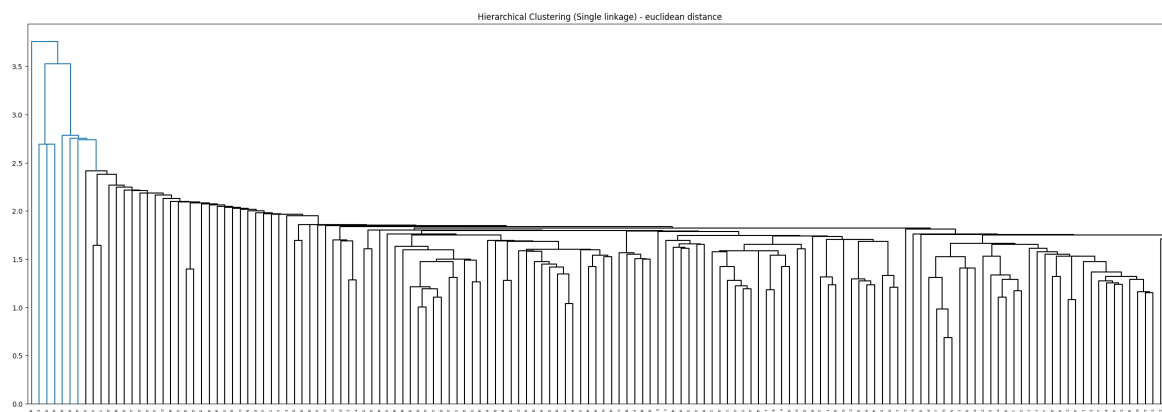


Figure 2.a.1: Hierarchical Clustering - Single Linkage, Euclidean Distance

Adjusted Rand Index (ARI): -0.010913368924245862

Adjusted Mutual Information (AMI): 0.013447552615202992

Manhattan Distance

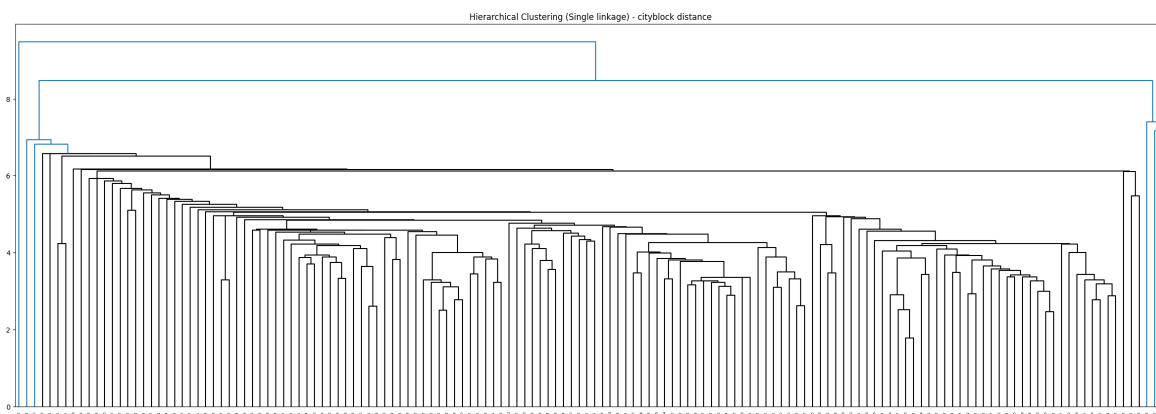


Figure 2.a.2: Hierarchical Clustering - Single Linkage, Manhattan Distance

Adjusted Rand Index (ARI): -0.010913368924245862

Adjusted Mutual Information (AMI): 0.013447552615202992

Complete Linkage

Euclidean Distance

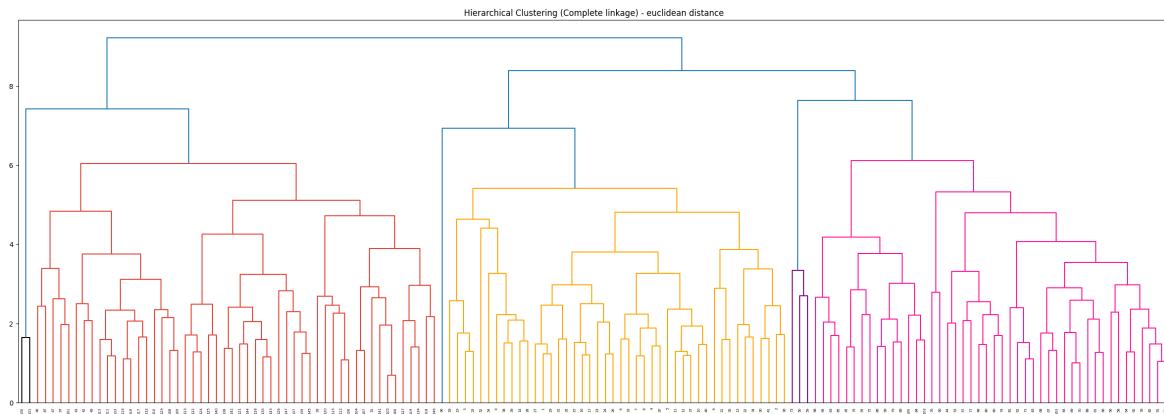


Figure 2.b.1: Hierarchical Clustering - Complete Linkage, Euclidean Distance

Adjusted Rand Index (ARI): 0.745740896797098

Adjusted Mutual Information (AMI): 0.7620088581447081

Manhattan Distance

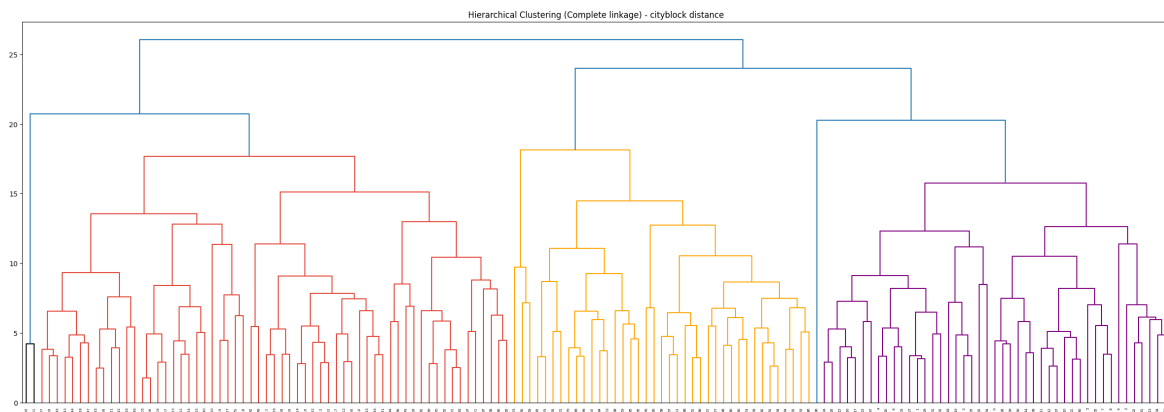


Figure 2.b.2: Hierarchical Clustering - Complete Linkage, Manhattan Distance

Adjusted Rand Index (ARI): 0.5958407796009646

Adjusted Mutual Information (AMI): 0.6716204925118936

Conclusions

S.No	Linkage Metric	Distance Metric	ARI	AMI
1	Single Linkage	Euclidean Distance	-0.0109	0.0134
2	Single Linkage	Manhattan Distance	-0.0109	0.0134
3	Complete Linkage	Euclidean Distance	0.7457	0.7620
4	Complete Linkage	Manhattan Distance	0.5958	0.6716

Single linkage performs the worst among the linkage methods, as illustrated by Figures 2.a.1 and 2.a.2, (This is due to its tendency to form a single, large cluster, failing to produce meaningful groups), as well as the consistently low ARI and AMI scores regardless of the distance metric used, as shown in the table above.

Complete linkage performs well, and as shown in the table above, the use of the Euclidean distance metric yields far better results compared to the Manhattan distance metric.

Thus, Complete linkage with the Euclidean distance metric performs best for this dataset and, in terms of ARI and AMI scores, it outperforms even the Ward linkage (with the Euclidean distance metric) from Question 1.

Question 3

Silhouette Coefficient based Analysis:

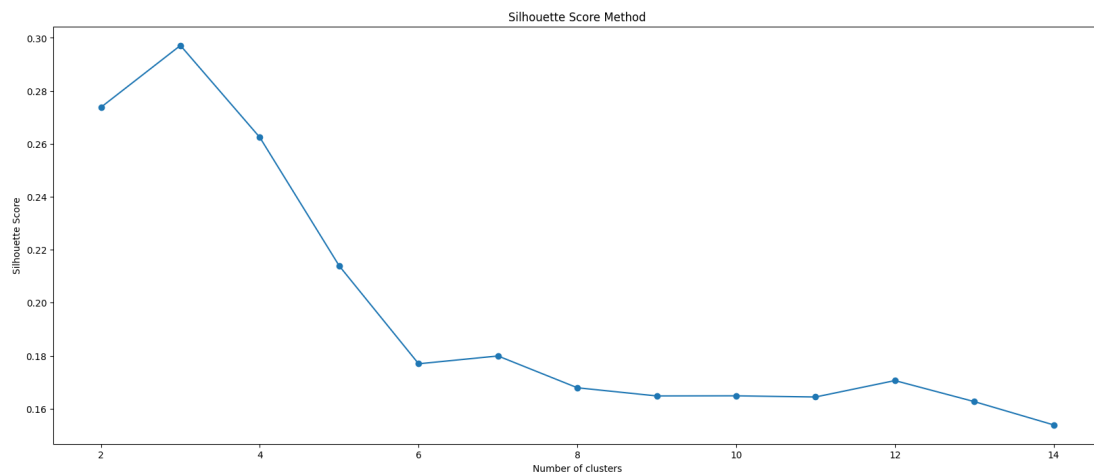


Figure 3.1: Silhouette Coefficient vs k values

Silhouette Coefficient: Is a measure of how well a data point is assigned to its cluster compared to its distance to points in other clusters. The optimal k is that which has the largest Silhouette Coefficient. From Figure 3.1 it becomes evident, that $k\text{-optimal} = 3$.

Calinski-Harabasz Index based Analysis:

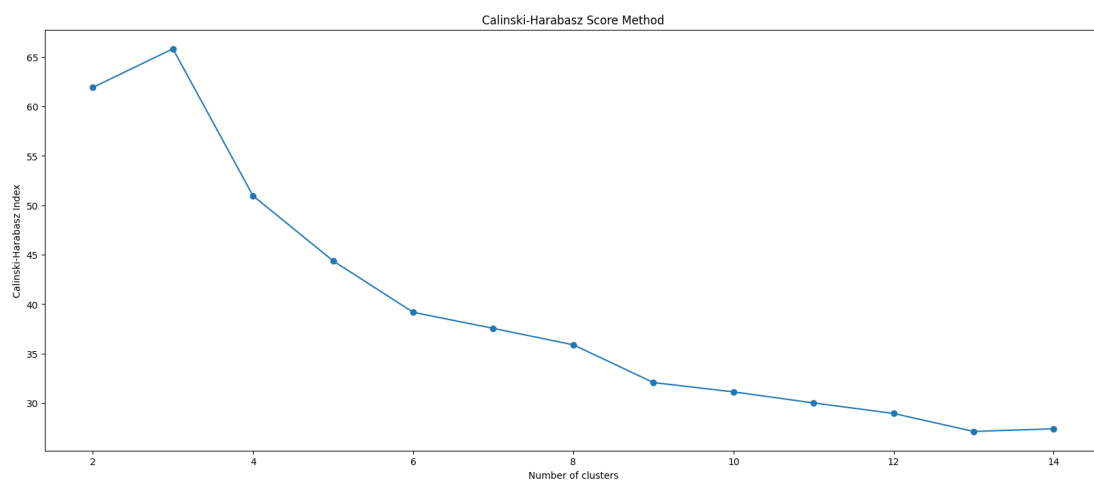


Figure 3.2: Calinski-Harabasz Index vs k values

Calinski-Harabasz Index (CHI): The CH Index is a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation). A larger value of CHI indicates that the data-points are well-matched to their clusters. The optimal k is that which has the largest Calinski-Harabasz Index. From Figure 3.2 it becomes evident, that $k\text{-optimal} = 3$.

NOTE: In a strict sense, we shouldn't be calling the above plots "elbow plots", as the selection criterion is not based on the point where the slope changes drastically (Here we choose the k corresponding to the maximum value of the metric used in both cases). The term "elbow plot" is more appropriately used when using evaluation metrics such as inertia.

Question 4

K-Means clustering was performed with $k = 3$, as determined from Question 3. The ARI, AMI scores and the output (cluster) labels, are given below:

[illegible]

Compared to agglomerative hierarchical clustering (using Ward or Complete linkage with the Euclidean distance metric), K-Means clustering with $k = 3$ achieves much better performance, indicated by the higher ARI and AMI scores.

The improved performance of K-Means clustering might be indicating that the clusters in the current dataset are radially symmetric in the 11-dimensional feature space, which also hints at why Euclidean distance in Question 2 performs better.

A notable difference between K-Means and the previous methods is that it clusters groups 0 and 1 with relatively few errors. However, it performs poorly on group 2, with many samples misclassified into clusters corresponding to groups 0 and 1. This is different from the Agglomerative clustering technique which had difficulty properly classifying group 1.

Question 5

We use DBSCAN (with $\epsilon = 2.2$) to determine the outliers in the given data. The number of outliers detected, along with the corresponding evaluation metrics (ARI, AMI) before and after their removal, are given below:

```
Unique clusters: [-1  0  1  2]
Number of outliers: 35
Number of non-outliers: 113
```

```
ARI (including outliers): 0.6314
AMI (including outliers): 0.6923
=====
ARI (excluding outliers): 0.9738
AMI (excluding outliers): 0.9622
```

The above results indicate that the removal of outliers leads to a clear improvement in model performance. Now, using this "clean" dataset, we repeat the analyses executed in Questions 1 and 4 to see if there is any improvement.

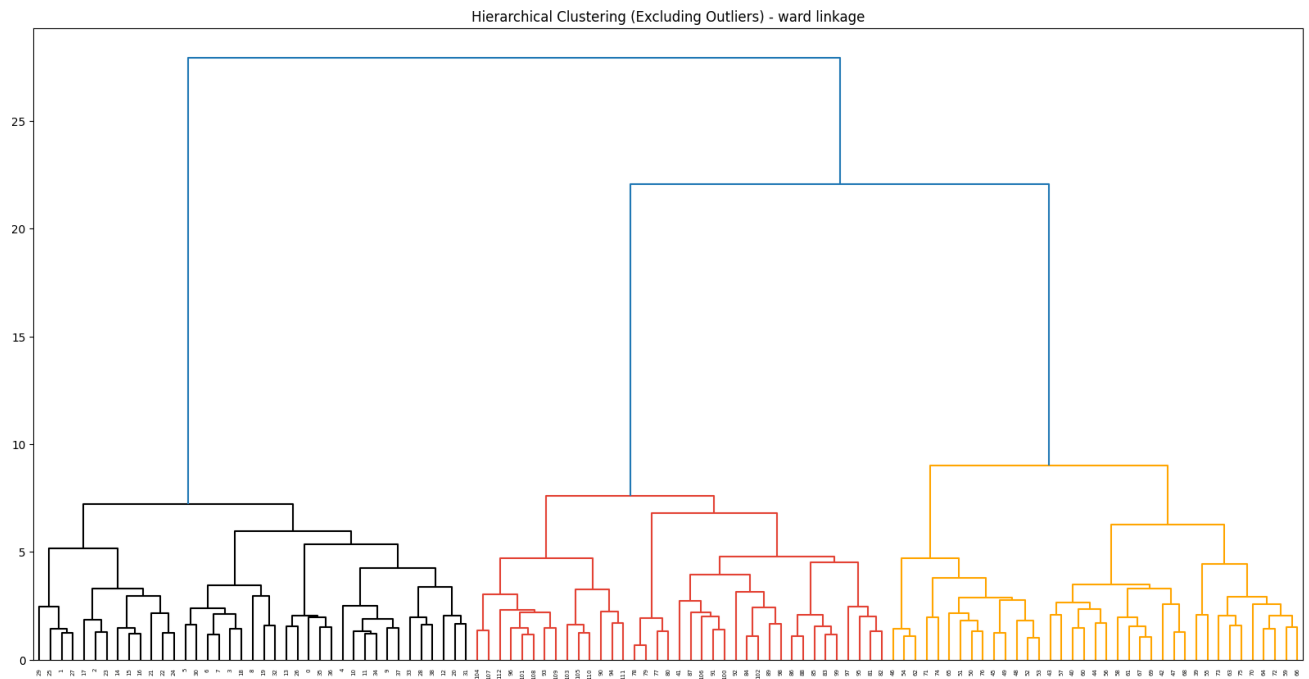


Figure 5.a.1: Agglomerative Hierarchical Clustering (Dendrogram) - Ward Linkage

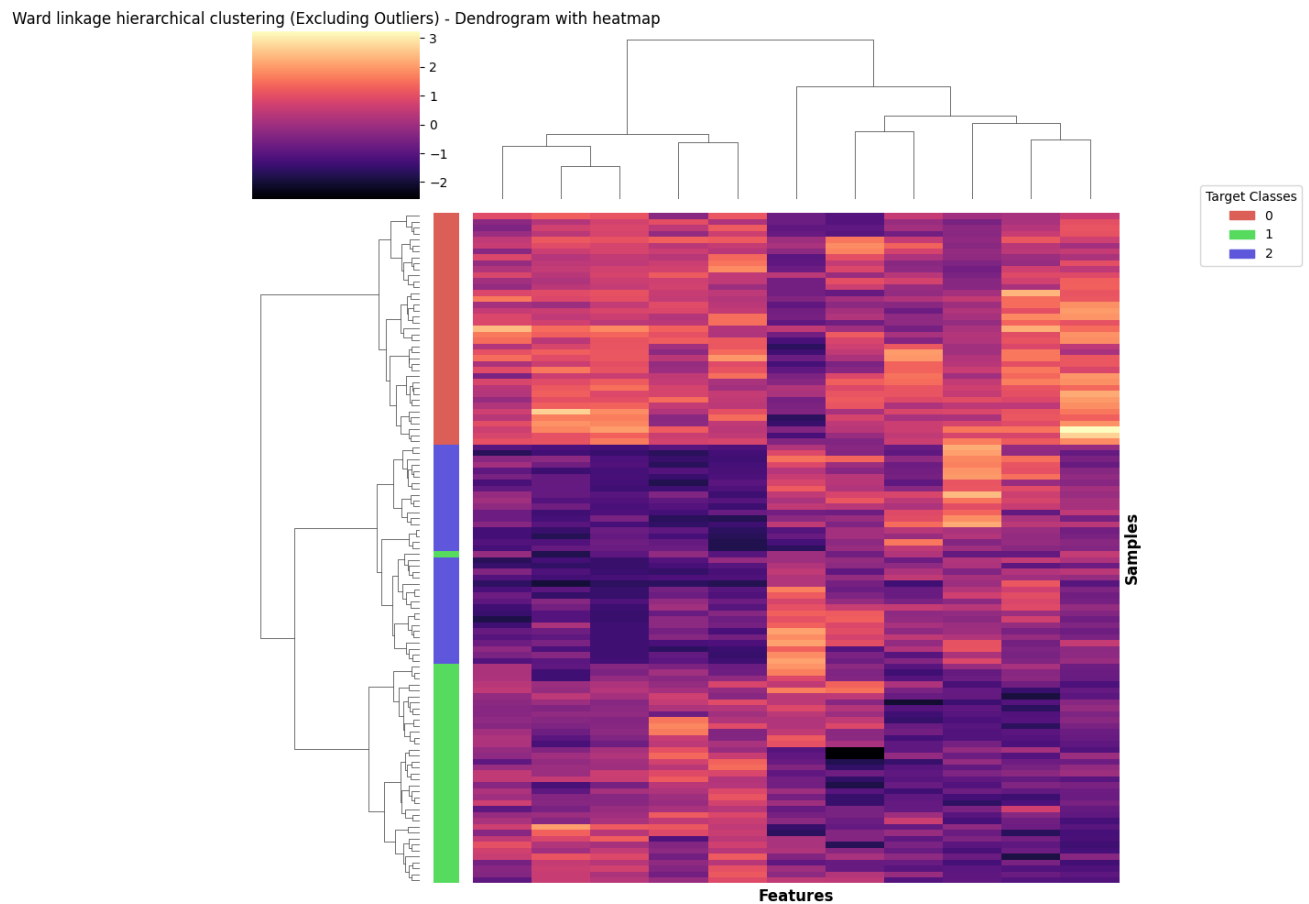


Figure 5.a.2: Ward linkage Agglomerative Hierarchical Clustering - Dendrogram with Heatmap

Agglomerative Hierarchical Clustering (Ward linkage)

ARI (excluding outliers): 0.9738
AMI (excluding outliers): 0.9622

The high ARI score indicates that most samples are clustered in similar to their true labels, while the high AMI score further suggests that the model effectively captures the underlying ground-truth accurately.

From Figure 5.a.1, it is observed that the clusters are overall closer to each other than before, with the maximum separation between clusters being ≈ 8 (as compared to 13 before).

Additionally, from Figure 5.a.2, we see that nearly all the samples are correctly clustered, with only one group 1 sample being misclassified as group 2. The heatmap also exhibits clear distinction in the feature patterns across clusters.

When comparing with the results obtained from Question 1, we can say that the current model demonstrates an improved ability to capture the ground truth and produces more accurate predictions (Much larger ARI and AMI values).

K-Means Clustering (k-optimal = 3)

K-Means clustering was performed with $k = 3$ on the "clean" dataset. The ARI, AMI scores and the output (cluster) labels, are given below:

[illegible]

The high ARI score once again indicates that most samples are clustered in similar to their true labels, while the high AMI score further suggests that the model effectively captures the underlying ground-truth accurately.

It is crucial to note, however, that the ARI and AMI scores for K-Means clustering are lower than those obtained from agglomerative clustering, indicating that agglomerative clustering might be better suited for this dataset (The "clean" dataset is most likely not radially symmetric in the 11 dimensional feature space).

We can see that a large chunk of the dataset has been classified accurately, with only a single group 2 sample misclassified into group 1.

(**NOTE:** 2 - Group 0, 0 - Group 1, 1 - Group 2 in Hierarchical clustering/The original data)

When compared to the results obtained in Question 4, there are significantly fewer misclassifications. Additionally, the greater ARI and AMI values indicate that the current model captures the ground truth more effectively.

In both cases the model accuracy rises which clearly indicates that the clustering performance was previously limited by noise/outliers and only minimally influenced by choice of approach.

Link to the google drive containing the python code used in this assignment:
<https://drive.google.com/drive/u/0/folders/1nkI4hVhCGi0d5xevm1o11bo65i-9SYA9>